

Revised Syllabus

Course Information

Item	Information
Course	UFOR 271 -- Environmental Science (From the Bluff to the Bayou)
Credit / contact	3 credit hours · Lecture 2 hrs · Laboratory/field 2 hrs
Prerequisite	SBIO 101B
Computing environment	ACCESS Cloudbank Classroom (browser-based Jupyter Hub) no local install
Term / format / meeting	[Term & year] · [In person / hybrid] · [Days, times, room, lab/field location]
Instructor	Dr. Tanganika Johnson

Student Learning Outcomes

On completion of the course, students will be able to:

1. Explain how ecological, geological, atmospheric, and hydrological processes interact within environmental systems.
2. Analyze relationships among human populations, natural resources, environmental quality, and sustainability.
3. Collect, document, manage, visualize, and interpret environmental field and secondary data.
4. Evaluate environmental hazards using scientific evidence, exposure pathways, dose-response concepts, uncertainty, and risk.
5. Examine how environmental burdens and benefits are distributed across communities.
6. Apply environmental ethics, economics, laws, and policies to authentic environmental decisions.
7. Use geospatial data, public environmental databases, and community-science tools to investigate environmental patterns.
8. **Use reproducible, browser-based computational tools (Jupyter notebooks on a science gateway) to visualize and analyze authentic environmental data.**
9. Design and conduct a collaborative environmental research project using responsible and reproducible practices.
10. **Apply responsible, transparent artificial-intelligence practices — disclosing, verifying, and appropriately limiting AI use in scientific work.**
11. Communicate environmental findings to scientific, professional, and community audiences.
12. Propose an evidence-based and equitable response to a South Louisiana environmental challenge.

Technology & Computing Environment

All computational work runs in the browser on **ACCESS CloudBank Classroom**, a managed teaching JupyterHub. Students sign in with their Southern University credentials; each receives persistent storage and a preconfigured Python environment, so there is nothing to install and every student works on equal

footing. Notebooks are distributed with **nbgitpuller** (one click) and are fully templated: students run cells and change clearly labeled values rather than writing code from scratch.

Data and tools come from open science gateways, selected to fit the CURE themes:

- **OpenTopography** : elevation and LiDAR data for terrain, coastal, and urban context.
- **iNaturalist** and **EPA EnviroAtlas** : biodiversity observations, ecosystem-service and environmental-justice metrics.
- **geemap** / **leafmap** : interactive geospatial mapping in Jupyter (leafmap requires no external account, keeping the barrier low).
- **HazMapper** (DesignSafe) : organizing geotagged field observations into a shared map.

Equity safeguard: every gateway activity has a spreadsheet / CSV fallback that meets the same outcome, so no student is blocked by a device or a slow connection.

The Jaguar Environmental Resilience CURE & Four-Stage Research Workflow

Teams choose one environmental-resilience theme (for example urban heat, water quality, or biodiversity) and carry it through a four-stage workflow that mirrors authentic research. The same scaffold applies to any theme.

Stage	What students do	Environment
Discovery	Locate and document authentic datasets; evaluate provenance, quality, and limitations; frame the research question	Browser gateways (EnviroAtlas, iNaturalist)
Visualization	Map and plot the data; describe the pattern they see	Templated Jupyter notebook (geemap / leafmap)
Analysis	Quantify relationships; treat uncertainty honestly	Jupyter notebook on CloudBank
Interpretation	Interpret through an environmental-justice lens; recommend an equitable, evidence-based response	Report, poster, community product

A short **HPC Carpentry “Introduction to HPC”** orientation opens the computational strand — for shared vocabulary and comfort, not heavy computing.

How AI, HPC, and Gateways Are Used selectively

Computational tools are applied where they deepen learning and left aside where they would only add friction. The learning outcome is hands-on investigation; the tools serve it.

Topic / activity	HPC / compute	AI	Kept at
Urban heat / tree canopy (satellite & LiDAR layers)	browser notebook handles large rasters	Limited, (debug a cell, suggest a chart)	Templated notebook, intro level
Coastal land loss / elevation (LiDAR)	OpenTopography + guided notebook	Limited	Guided tutorial
Water quality (large / multi-year records)	Only if it outgrows a spreadsheet	Limited (formula help)	Spreadsheet first

Topic / activity	HPC / compute	AI	Kept at
Biodiversity counts (iNaturalist)	Not needed	Limited (brainstorm and researching)	Browser, no compute
Field journal / observations	Not needed	Avoid (data must be real & student-collected)	Browser + HazMapper
Environmental-justice interpretation	Not needed	Avoid for interpretation → no hallucination	Human-led; disclosure only
Concepts, ethics, policy, community product	Not needed	Permitted for clarity/tutoring, with verification	Low-tech / writing

Artificial-Intelligence Use Policy

AI is a workplace tool that can support environmental data work while also introducing errors, bias, confidentiality concerns, and resource impacts. **Every assignment is labeled AI prohibited, AI limited, or AI permitted.**

Tier	Meaning
AI prohibited	Complete the work without generative AI so independent reading, calculation, observation, reasoning, or reflection can be assessed.
AI limited	AI may support only the listed tasks (e.g., brainstorming search terms, checking code syntax, improving clarity); it may not replace source reading, data analysis, or interpretation.
AI permitted	AI may be used as directed, with verification, disclosure, appropriate citation, and required process evidence.

How AI is implemented in this course

- **Where AI helps (permitted / limited):** brainstorming search terms; debugging or explaining a notebook cell; suggesting a chart type; improving the clarity of writing; summarizing literature in the source-review milestone — with every citation verified against the original.
- **Where AI is not used:** generating or altering field observations, measurements, images, or data (these must be real and student-collected); interpreting environmental-justice or community narratives (kept human, because AI can flatten community context); and any factual claim that has not been verified against authoritative sources.
- **Disclosure:** when AI use is allowed, students briefly state the tool, the purpose, the prompts or process when requested, and how the output was verified or changed. Major products include a short **AI-use appendix**.
- **Integrity:** assessments that generative AI could otherwise complete (the Decision-Making Case and the Final Environmental Solutions Assessment) are evaluated on documented process, validation, and a brief oral defense.
- **Responsibility:** students remain accountable for accuracy, bias, source quality, confidentiality, copyright, data integrity, and the environmental cost of tool use. No personal or community data is entered into AI tools. AI output is not a credible scientific source by itself.

Course Modules

The course retains its five modules and the Engage → Learn → Investigate → Analyze → Act/reflect cycle; the gateway and notebook activities are woven into the Investigate and Analyze phases.

- **M1 Reading the Environment** (Wks 1–3): systems, data literacy, measurement, public databases. *CloudBank onboarding + HPC Carpentry orientation; CURE question.*
- **M2 People, Place & Environmental Health** (Wks 4–6): population, environmental justice, toxicology, risk. *EnviroAtlas discovery activity; Decision-Making Case.*
- **M3 Water, Wetlands & the Working Coast** (Wks 7–9): hydrology, wetlands, coastal land loss. *OpenTopography / mapping activity; CURE methods & data collection.*
- **M4 Energy, Climate & Resilient Communities** (Wks 10–12): energy, climate, heat, resilience. *Notebook analysis (e.g., urban heat × canopy); Environmental Systems Exam.*
- **M5 Environmental Decisions & Community Action** (Wks 13–16): ethics, law, policy, communication. *Interpretation & community product; research showcase; Solutions Final.*

Assessment

The course uses a 500-point system (40% formative / 60% summative), retained from the base course with the CURE gateway activities embedded in the formative milestones and AI-integrity elements added to the summative.

Category	Components	Points
Formative (200)	OER reading checks; environmental field journal; four environmental-data / gateway activities (the four-stage workflow); five CURE milestones; participation	200
Summative (300)	Environmental Systems Exam (50); Decision-Making Case + AI-use appendix (50); CURE Research Report (75); Scientific Poster & Presentation (50); Community Environmental Product (25); Final Environmental Solutions Assessment (50)	300

Grading scale: A 450–500 · B 400–449 · C 350–399 · D 300–349 · F below 300.

Technology & Research Activity Schedule

Weeks	Computational / gateway / AI activity	AI tier
1–2	CloudBank onboarding; HPC Carpentry “Intro to HPC” orientation; first templated notebook run	Limited
3	Data-discovery activity in EPA EnviroAtlas; frame CURE question	Limited
4–6	Environmental-justice pattern activity; Decision-Making Case (with AI-use appendix)	Limited / disclosed
7–9	OpenTopography / mapping activity; CURE data collection (HazMapper for field data)	Limited
10–12	Notebook analysis of the chosen theme (visualization + analysis stages)	Limited
13–16	Interpretation, poster, community product; research showcase; Solutions Final	Permitted (clarity) / disclosed

This schedule layers onto the retained 16-week content calendar; field activities may be rescheduled for weather, air quality, transportation, accessibility, or safety.

Academic Integrity, Accessibility & Data Ethics

- **Academic integrity & source use:** all borrowed material, media, code, and datasets are cited in the course-approved style; fabricated sources, falsified or selectively omitted data, unauthorized collaboration, undisclosed AI-generated work, and plagiarism are addressed under Southern University procedures.
- **Accessibility & inclusive design:** materials follow WCAG 2.2-informed practices (headings, descriptive links, contrast, alt text, captions, accessible tables); browser-based tools and non-visual data paths keep activities usable; accommodations are arranged through the Southern University accessibility office.
- **Data ethics & reproducibility:** preserve original observations; document cleaning, exclusions, and transformations; use descriptive filenames, metadata, and units; do not publish sensitive locations or personal/community data; report uncertainty and null results honestly; follow every dataset's license and terms.
- **Field & research safety:** required safety preparation before field/lab work; no entry to private property, hazardous sites, or moving water; no collection of people's images or personal data without instructor-approved protocol; accessible, weather-safe alternatives assess the same outcomes.

Items to Finalize Before Publication

- Instructor contact, term, meeting information, delivery format, and office hours.
- Current Southern University required policy statements and student-support links (integrity, accessibility, Title IX, emergency, counseling, basic needs).
- Confirmed CloudBank Classroom allocation and student-account process; ACCESS onboarding schedule.
- Exact readings and open-license confirmation; due dates, holidays, and late-work window; approved citation style and templates.
- CURE theme, sites, equipment, transportation, safety, and data-management plan; gateway contingency resources.